I. Data to Be Collected

a. Goal

Mission Objective: Autonomously characterize Europa's ice shell thickness along a surface traverse and identify an optimal cryobot insertion site without real-time Earth support (REQ.03.01: ground-penetrating radar sounder resolves ice thickness to $\pm 10\%$ for depths of 1–30 km at ≥ 100 m along-track spatial resolution).

Data Product Goal: Provide engineers and scientists with a time-aligned record of radar sounder returns, rover localization, terrain context, and system health to verify that the subsurface map meets REQ.03.01 and that the onboard site-selection algorithm correctly identifies a candidate cryobot insertion point.

Success Criterion: Full traverse track covered at ≤ 100 m along-track spacing, ice thickness accuracy within $\pm 10\%$ against calibration targets, and at least one candidate insertion site flagged by the onboard algorithm (ice thickness <5 km, surface slope $<20°$, terrain roughness <0.7).

b. Data Collected by the Robot

- **Category 1 — Radar Sounder (Primary Science):**
  - Two-way travel time to basal reflector (ns)
  - Received signal amplitude at basal reflector (dB)
  - Estimated bulk ice dielectric permittivity (dimensionless)
  - Derived ice shell thickness (m)
- **Category 2 — Rover Localization:**
  - Position estimate from visual odometry and inter-agent ranging (m, local ENU)
  - 1σ position uncertainty (m)
  - Rover heading (deg)
  - Localization error flag (Boolean: uncertainty exceeds ± 0.5 m per REQ.02.01)
- **Category 3 — Terrain Context:**
  - Surface slope at measurement site (deg)
  - Terrain roughness index (dimensionless, 0–1)
  - Surface temperature (K)
- **Category 4 — System Health:**
  - Total ionizing dose accumulated by radar electronics (krad)
  - Radar noise floor (dB)
  - Inter-agent link packet loss rate (%)
  - Per-trace data quality flag (NOMINAL / CLUTTER / NOISE_FLOOR / INVALID)
- **Daily Summary:**
  - Total traverse distance completed in sol (m)
  - Valid radar trace count (count)

- Minimum observed ice thickness in sol (m)
- Candidate insertion sites flagged by onboard algorithm (count)
- Mean inter-agent consensus convergence time (min)

II. Types of Measurements on the Spreadsheet

The spreadsheet is organized into multiple sheets to keep the dataset clear and usable. Below is a breakdown of each sheet:

1. **Sheet 1 — Radar_Traces:**
   Time-aligned record of all sounder measurements, one row per trace. Records the along-track position, two-way travel time, received amplitude, estimated permittivity, and derived ice thickness (computed as $d = c \cdot t/2\sqrt{\varepsilon_r}$), alongside a per-trace quality flag. This is the primary science record that feeds the subsurface map and site-selection logic.

2. **Sheet 2 — Localization_Log:**
   Per-trace rover position fix synchronized to the Radar_Traces timestamp index. Captures position (X, Y), heading, 1σ uncertainty, and the REQ.02.01 error flag. Georeferences each sounder trace to a physical surface coordinate.

3. **Sheet 3 — Terrain_and_Health:**
   Per-trace terrain and system health data, aligned to the same trace index. Captures surface slope, roughness, temperature, accumulated TID, radar noise floor, and inter-agent packet loss. Contextualizes each radar measurement and flags traces collected under degraded system or terrain conditions.

4. **Sheet 4 — Daily Summary:**
   One row per sol. Auto-populated via formulas from Sheets 1–3: total traverse distance, valid trace count, minimum ice thickness, candidate site count, and mean consensus convergence time. Primary operational report for mission planning and onboard replan decisions.

IV. Rationale and Alignment with Mission Objective

Alignment with Objective: Each data category maps to a specific requirement. The radar sounder returns (Category 1) are the direct measurement required by REQ.03.01. Without travel time and permittivity there is no thickness inversion and no subsurface map. The localization log (Category 2) is required by REQ.02.01: each trace must be placed within ± 0.5 m to produce a spatially accurate map rather than an unordered list of depth readings. Terrain context (Category 3) drives the site-selection filter, where slope and roughness constrain physical accessibility independent of ice thickness. System health data (Category 4) is required by REQ.01.02 and REQ.04.01: elevated TID can degrade radar performance, and packet loss above 30% directly stresses the consensus convergence bound.

Why These Measurements Are Sufficient: The four categories together close the loop on REQ.03.01 and its dependencies. Travel time plus permittivity produces thickness; localization georeferences it; terrain context filters by physical accessibility; health flags bound measurement confidence. The daily summary aggregates these into the per-sol statistics the onboard planner needs to decide whether to continue traversing or commit to a cryobot insertion attempt.

Potential Gaps or Limitations: The main limitation is permittivity estimation uncertainty when brine inclusions or heterogeneous layering are present (RISK-05 in the requirements document). Brine lenses increase local permittivity and produce reflections that can cause the thickness algorithm to underestimate depth to the basal interface. The CLUTTER quality flag mitigates this by excluding affected traces from site selection, but does not resolve the ambiguity without a secondary instrument. Additionally, the along-track resolution of 100 m defined by REQ.03.01 produces a 1D profile along the traverse; cross-track coverage depends on orbital relay scan width, which is not yet specified.

AI Disclosure: Artificial intelligence tools were used to verify that the technical statements, parameter values, and requirement references made in this document are consistent with the background research on Europa ice-shell radar sounding, cryobot mission concepts, and planetary robotics requirements engineering. AI also generated fake data to demonstrate the data being recorded.